Discussion Grades: setup guide

A live grading app for discussion sections: TAs enter grades on their phones, every grade lands in a Google Sheet as it's typed, and a menu turns a class day into a Canvas import file.

What it is. A Google Sheet with a small Apps Script web app attached. There's no server and nothing to pay for. It runs in your own UW Google account, so the grades stay with you. Setup takes about half an hour, and starting partway through the semester is fine.

In this kit	What it is
Setup guide.pdf	This guide.
Code.gs	The app's server side. You paste it into Google. There's nothing to change in it.
Index.html	The screen TAs see. You paste it into Google. There's nothing to change in it.
Settings.gs	Your course's settings. You paste it into Google once, then edit it. Updates never replace it.
Example roster.csv	48 made-up students in four sections. It shows what the Roster tab looks like once it's filled in. You don't need it for setup.

The newest version of every file is always at `github.com/carlybenish/discussion-grades`, along with a list of what's changed. Each file there has a **Copy raw file** button, which is the easiest way to copy it into Google.

> **Use your UW–Madison Google account (NetID@wisc.edu) for all of this, not a personal Gmail.** It keeps the grades in your UW account, and it's what lets you limit the app to UW people. In a browser with several Google accounts, check the picture in the top-right corner before each step.

PART 1 · SETUP

1 Make the spreadsheet

Go to **sheets.new** and name the blank sheet, e.g. `EMA 303 Live Grades`. Leave it empty: the roster goes in at step 3, straight from Canvas.

2 Add the app and set your course

1. In the spreadsheet, choose **Extensions** → **Apps Script**. A code editor opens in a new tab.
2. Copy all of **Code.gs**: on the GitHub page, click it, then **Copy raw file** (two overlapping squares, top right). Or open it from this kit in a text editor (on a Mac: right-click → Open With → TextEdit) and copy everything. Paste it over everything in the editor's `Code.gs`.
3. Click the **+** next to "Files", choose **HTML**, and name it exactly `Index` (Google adds the `.html`). Paste the whole of **Index.html** over what's there.
4. Click **+** again, choose **Script**, and name it `Settings`. Paste the whole of **Settings.gs** over what's there.
5. In `Settings`, edit the **YOUR COURSE** block (below), then click save 💾.

```
var COURSE = {
  name: 'Course 101',                                  // e.g. 'EMA 303'
  sections: ['DIS301', 'DIS302', 'DIS303', 'DIS304'],
  classDays: ['Wed', 'Fri'],
  firstDay: '2026-09-02',
  lastDay: '2026-12-11',
  maxGrade: 10,
  defaultGrade: 10,
  lateSteps: [[5, 9], [10, 8], [15, 7], [20, 6], [25, 5], [26, 4]]
};
```

Setting	What it does
name	Shown at the top of the app and on the page people see if they're not on the TA list.
sections	Your discussion sections, in the order the tabs should appear. A name like `DIS301` shows as "301" on the tabs. Each one must match the roster's Section column (`DIS301` matches `DIS301`, `301`, and Canvas's "…:DIS301 and …").
classDays	The weekdays sections meet, three letters each: `'Mon'` `'Tue'` `'Wed'` `'Thu'` `'Fri'`. The day picker offers every one of these from firstDay to lastDay, and opens on today.
firstDay, lastDay	The semester's first and last class days, written year-month-day, in quotes. The dates already there cover Fall 2026, so leave them for this semester. Starting mid-semester is fine: the app opens on today, and earlier days just stay empty.
maxGrade	Grades run from 0 to this (up to 99). Decimals like 7.5 are allowed.
defaultGrade	What tapping an empty box fills in (usually full credit).
lateSteps	The Late button's choices: `[up to this many minutes late, the most they can get]`, smallest first. The buttons show ranges: with the steps above, 1–5 min late can get at most 9, 6–10 min at most 8, and so on. The last step means "that late or later" (26+). To track lateness without taking points off, use `[[1, 10]]` (with a maxGrade of 10): Late becomes just a flag. TAs don't have to use the button at all.

Keep the quotes, commas and brackets as they are. If something in the block is off, the app's link shows a page saying which setting to fix instead of the app.

3 Load the roster from Canvas

1. In Canvas: **Grades** → **Export** → **Export Entire Gradebook**. A CSV file downloads. There's no need to open or edit it.
2. Go back to the spreadsheet's browser tab and reload it. After a few seconds a **Roster** menu appears next to Help.
3. Choose **Roster** → **Update from Canvas**. The first time, Google asks you to authorize the script: pick your UW account, and if it says "*Google hasn't verified this app*", click **Advanced** → **Go to (project name)** → **Allow**. That warning appears for any script you write yourself. Then choose the menu item again.
4. Pick the file you downloaded. The window says what it will do; click **Update the roster**. A **Roster** tab appears with every student, by section and then name. The Test Student is left out.

Roster column	Filled in by	What it's for

Section	Canvas	The student's discussion section. Students whose section isn't in your `sections` list stay on the tab but don't show in the app; the window lists them.
Student	Canvas	The name TAs see, as Canvas writes it: `Lastname, Firstname`.
Canvas ID, SIS User ID, SIS Login ID, Canvas section	Canvas	How the Canvas menu matches grades to students. Leave these alone.
Table, Group	You, if you want	If students sit at numbered tables in groups (Table `4`, Group `B`), fill these in and the app lists them that way, with a jump bar by table. Leave them blank and each section is one list.
TA (or Wed TA, Fri TA, …)	You, if you want	Add the column yourself: which TA covers that student's table, shown in the app as a label. Use one `TA` column, or one per class day named after the day (`Wed TA`). Also optional: `Lead` and `Support` (or `Wed Lead`, …), one name per section.

Already have a roster file with tables and groups? Import it instead (**File** → **Import** → **Upload**, **Replace spreadsheet**, then rename the tab to exactly **Roster**), then run **Roster** → **Update from Canvas**. It fills in the Canvas IDs, matching each row by name, so the names have to be written the way Canvas writes them. Compare with **Example roster.csv**.

About names

The app never splits a name into first and last names. TAs see the Student cell exactly as it's written, so four first names or two last names are fine, and a long name wraps onto a second line. Grades reach Canvas through the IDs on the student's row, not through the name, so you can shorten a name to the one the student goes by. Do it before the first day you grade, because each class day's tab keeps the names it was made with. If two students in one section have the same name, the import adds a login to tell them apart, e.g. `Lee, Sam (slee4)`.

4 Turn it on

1. **Deploy** → **New deployment**. Click the gear ⚙ next to "Select type" and choose **Web app**.
   - **Execute as:** Me
   - **Who has access:** Anyone within University of Wisconsin–Madison
2. Click **Deploy**. If Google asks you to authorize again, do it the way you did in step 3.
3. Copy the **Web app URL** (it ends in `/exec`) and **add `/a/macros/wisc.edu` to it**:

   ```
   Google gives you:  https://script.google.com/macros/s/AKfycb…/exec
   Give the TAs:      https://script.google.com/a/macros/wisc.edu/s/AKfycb…/exec
   ```

   It's the same app. The second form tells Google the link belongs to UW, so a TA whose browser is sitting in a personal Gmail gets an account picker instead of a dead-end "You need permission" page. Try it once in a private window before sending it.

> **Never set "Who has access" to "Anyone."** It would lock out everyone, TAs included: anonymous visitors have no identity for the app to check against the TA list (next step), so it turns them all away.

5 Say who the TAs are

"Anyone within UW–Madison" is about 100,000 accounts, including every student in your course, so that setting isn't what keeps the grades private. A **TA access** tab is: only addresses listed there can open the app. Anyone

else gets a polite "this app is for (your course) TAs" page with no student information on it.

1. Open the link yourself once. A **TA access** tab appears in the spreadsheet with your own address in it. (Reload the spreadsheet if the new menus aren't showing.)
2. Add one row per TA, with their email in column A. If your roster has TA columns, **TA access → Add the TAs from the roster** puts every TA's name on the tab with the email blank. Fill in each **yellow** row; yellow means that row lets nobody in yet.

Which address? The NetID one (`bbadger@wisc.edu`), not a vanity alias (`bucky.badger@wisc.edu`). Google hands the app the account's main address no matter which one the TA typed. When unsure, put both on two rows; a spare row can't let anyone else in. Or let the TA try the link: they're turned away once, and **TA access → Blocked sign-ins** shows the exact address the app saw. Paste that in and they're in within a minute.

Your own address always works, even if it's deleted from the tab, so you can't lock yourself out.

6 Test it, then hand it out

1. Open the link and type a grade for any student. In the spreadsheet, a tab named for that day (e.g. `Fri 9/11`) appears with the grade in it. Clear the grade in the app afterward.
2. Give the TAs the `/a/macros/wisc.edu/` link. **They don't need access to the spreadsheet.** Best: put it on the Canvas course page (e.g. a module item "Enter discussion grades"), so it lives in one place you can edit. Avoid URL shorteners: the free ones can't re-point a link later.
3. Have each TA open it once and add it to their phone's home screen (Safari: Share → Add to Home Screen; Chrome: ⋮ → Add to Home screen). It takes a minute at a TA meeting and saves the "what's the link?" messages.
4. Share the spreadsheet itself (the **Share** button) with whoever enters grades in Canvas. Sharing the sheet gives full access to the grades, whether or not that person is on the TA access tab.

7 Tell it your Canvas assignments

Each class day's grades go into one assignment that already exists in Canvas. The app never creates assignments.

1. In the spreadsheet: **Canvas → Show Canvas assignment list**. A **Canvas assignments** tab appears, listing every class day.
2. Export the Canvas gradebook again (or reuse the one from step 3). Its first row names each assignment with its Canvas ID in parentheses, like `Discussion 3 (4012345)`. Copy those names for your discussion assignments into **column E**, one per row, exactly as written.
3. For each class day, pick its assignment from the dropdown in **column B**. Leave days with no discussion blank, and write why in column C if you like ("No class").

PART 2 · USING IT

For TAs

- Open the link. It opens on today's class (or the most recent class day). Tap a section, then tap a student's box: it fills in full credit, and it's saved when they tap the next student or anywhere else. To give a different grade, just type it. On a laptop, Enter saves and moves to the next student.
- **Skipping someone** (absent, not graded yet): don't tap their box. Moving onto a box with Enter or the arrow keys shows only a faint number, which isn't a grade, and moving on leaves it empty. Tapped the wrong student? Delete the grade before moving on.

- **Only a dark number with a green ✓ is saved.** If anything hasn't reached the spreadsheet, a red bar appears at the top and the row says "Not saved yet." The app keeps retrying; keep the page open until the bar goes away. On an iPhone, anything unsaved stays on the phone and goes through the next time the link is opened.
- **Late:** tap Late next to a name, then how many minutes. That sets the most the student can get, and fills it in if they had no grade yet. Tap Late again to clear it.
- **Absent:** tap Absent next to a name to give a 0 (the name is struck through, and a Late flag comes off). Tap it again to clear the 0.
- **Same for all** (the dashed box at the top of a group) gives the whole group one grade.
- **Someone's not on the list:** "+ Student not on this list" at the bottom of the section.
- **All** shows how far along every section is; tap a section to see only the students still missing a grade. Two TAs in one section see each other's grades within a few seconds.

In the spreadsheet

- Each class day gets its own tab, newest first, created the first time someone enters a grade that day. Yellow rows have no grade yet.
- The **Change log** tab records every change: old grade, new grade, who, and when. A grade that gets overwritten can always be recovered.
- A grade fixed directly in the spreadsheet shows up in the app within about 20 seconds.

Getting a day's grades into Canvas

1. Click the tab for the class day, e.g. `Fri 9/11`.
2. **Canvas → Make Canvas import from this day**. (The first time, Google asks you to authorize it, the same way as in setup.) It shows which assignment the grades go into. Click **Yes**.
3. A **Canvas import** tab appears. **File → Download → Comma-separated values (.csv)**.
4. In Canvas: **Grades → Import**, choose that file, **Upload Data**, and confirm the changes it shows.

- Students with no grade that day are left out of the file, so Canvas leaves them alone. It never enters a 0.
- A student the menu can't match to a Roster row (say, a walk-in who isn't on the Roster tab) goes on an **Enter by hand** tab instead. Enter those in Canvas yourself and tick **Entered in Canvas**.
- If two days point to the same assignment, the menu warns you before making the file.
- The **Export log** tab (Canvas → Export log) records who made each file and when.

When enrollment changes

Export the gradebook from Canvas again and choose **Roster → Update from Canvas**. It lists what it will change before it changes anything:

- **New students** go in under their section, in name order, highlighted yellow. Give them a table and group if you use them.
- **Section changes** move the student to the new section and highlight them. Their table and group come along, so check those.
- **Students no longer in Canvas** are listed and removed only if you tick the box. Their grades on past days stay.
- **Name changes** in Canvas are listed but not made, because past days' tabs use the Roster's names.

Everything else on the tab stays put, including the Table, Group and TA columns and any highlighting of your own. You can still edit the Roster tab by hand too; keep the headers as they are. Open copies of the app pick up a change within about 5 minutes, and reloading shows it sooner. Tabs for past days aren't changed.

Updates, and changing your settings

When the list of changes at `github.com/carlybenish/discussion-grades` has something new: open **Code.gs** there, click **Copy raw file**, and paste it over everything in your Apps Script `Code.gs`. Do the same with **Index.html** and your `Index` file. Leave `Settings` alone: updates never replace it. To change a setting instead, edit `Settings`.

Either way, save, then **Deploy** → **Manage deployments**, click the pencil ✏️, set **Version** to *New version*, and **Deploy**. The TA link stays the same. Don't use "New deployment" for this: it makes a second app with a different link. Reload the spreadsheet afterward so its menus pick up the change.

Finding the link again

Spreadsheet → **Extensions** → **Apps Script** → **Deploy** → **Manage deployments**, and copy the Web app URL. Add `/a/macros/wisc.edu` as in step 4 before sending it. A link ending in `/dev` is a test link that only works for you.

PART 3 · TROUBLESHOOTING

What happens	Why, and the fix
No "Anyone within University of Wisconsin–Madison" option	You're signed in to a personal Google account. Start over in your NetID@wisc.edu account.
A TA sees "You need permission"	Their browser's default Google account is a personal one. Send them the `/a/macros/wisc.edu/` form of the link; its account picker has "Use another account" for their NetID. A private window works too.
A TA sees "This app is for … TAs"	They reached the app but aren't on the TA access tab. The page names the account they're signed in as. If it's their wisc.edu address, add it to the tab and have them reload a minute later. If it's a personal address, see the row above.
"The course settings (Settings.gs) need fixing"	It says which setting. Fix it in `Settings`, save, then Manage deployments → ✏️ → New version → Deploy. If it says there's no settings file, add one as in step 2.
Update from Canvas says the file doesn't look like a gradebook export	Pick the file just as Canvas downloaded it (**Grades** → **Export** → **Export Entire Gradebook**). A file opened and re-saved in Excel can come out in another format, with accented names garbled.
Update from Canvas says none of the students are on the Roster tab	The file is probably another course's gradebook, so nothing was changed. If you typed the Roster tab in by hand, its names have to match Canvas's exactly (`Lastname, Firstname`).
"Couldn't load the roster"	The roster tab must be named exactly `Roster`, with the headers in row 1, including Section and Student.
A section is empty in the app	Its roster rows don't match a name in `sections`. Check the spelling in both places.
A TA says grades they entered aren't in the sheet	Check the Change log for those students. Nothing there means the grade never left their device: have them open the link again on the same device, which sends anything it kept.
The Roster, Canvas or TA access menu is missing	Reload the spreadsheet; the menus appear a few seconds after it opens.